STAFF REVIEW · H446 189 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Sorted input.

2. O(n log n).

3. 5 times.

4. 16 operations.

I Do — source worked steps

1. Predict first — Assume one successful or unsuccessful lookup in sorted data.

2. Model solution — Linear search can inspect all 16 elements; binary search checks at most 5 midpoints using this inclusive implementation.

We Do — guided CHECK

1. Model solution
Linear worst case = 64 comparisons; binary worst case at most 7. Binary grows much more slowly.

You Do — source mark schemes

1. O(n).

2. Linear up to 128 checks; binary up to 8 checks with a standard midpoint search.

3. No. An unsorted tiny array can be searched linearly without sorting. Binary search first needs sorted data, and sorting/preparation may cost more than one scan.

Exit Ticket — source answer

1. False. It describes the growth of operations, not a time in seconds.